THE CONSULTANT OF THE PROPOSED BUILDING WORKS AT PLOT NO (   ) BLOCK(   )
SHALL OBSERVE ADHERENCE TO THE GENERAL PROVISION OF THE BUILDING REGULATION AND COMPLIANCE WITH THE FOLLOWING NOTES :

STANDARD NOTES FOR BUILDING DRAINAGE

A) DOCUMENTATION

☐ A-1 DM DRAINAGE DEPARTMENT N O C NO: DATE
☐ A-2 DM DRAINAGE DEPARTMENT 'F I C DATE
☐ A-3 RECEIPT NO FOR SEW CONNECTION TOTAL BUILT UP AREA = SQFT.
☐ A-4 RECEIPT NO FOR STORM WATER CONNECTION
☐ A-5 TOTAL NO OF SEPTIC TANK
☐ A-6 TOTAL NO OF DM SEWERAGE . CONNECTION
☐ A-7 TOTAL NO OF DM STORM WATER . CONNECTION
☐ A-8 LAST BUILDING PERMIT NO: DATE

B) PIPING

☐ 1- MATERIALS SHALL BE TO DM APPROVAL AND THE INSTALLATION SHALL BE ACCORDING TO BS 5572
☐ 2- ALL INTERNAL PIPING SHALL BE UPVC TO BS 4514 EXCEPT WASTE PIPING WHICH SHALL BE TO BS 5255. ALL U/G PIPING SHALL BE TO BS 4660
☐ 3- NO BENDS SHALL BE ALLOWED IN HORIZONTAL PIPES RUNNING UNDERGROUND.
☐ 4- ACCOUSTIC INSULATION SHALL BE CONSIDERED FOR ALL DRAINAGE PIPES AND STACKS.

RODDING EYES

☐ 1- ACCESS POINTS SHOULD BE CAREFULLY SITED TO ALLOW THE SERVICE ENTRY FOR CLEANING AND TESTING .
☐ 2- ALL VERTICAL STACKS SHALL BE PROVIDED WITH RODDING EYES AT JUNCTIONS ON EVERY FLOOR.
☐ 3- ALL HORIZONTAL DRAINAGE PIPES SHALL HAVE RODDING EYES AT POINTS OF CHANGE OF DIRECTION IN PIPES AS REQUIRED FOR EFFECTIVE MAINTENANCE
☐ 4- RODDING EYES SHALL BE PROVIDED WHERE MORE THAN ONE WC IS CONNECTED TO A HORIZONTAL SOIL PIPE
☐ 5- RODDING EYE IS TO BE PROVIDED WHERE THE DISTANCE BETWEEN ONE WC AND STACK OR MANHOLE IS MORE THAN 5 METERS.

STACKS.

☐ 1- THE UPPER LIMIT OF STACK LOADING SHALL NOT BE MORE THAN A QUARTER FULL
☐ 2- FOR BUILDINGS OF LESS THAN 20 STORIES HEIGHT, APPLIANCES LOCATED ON GR. FLOOR SHALL NOT BE CONNECTED TO VERTICAL STACK DISCHARGING AT GR. LEVEL
☐ 3- FOR BUILDINGS GREATER THAN 20 STORIES APPLIANCES LOCATED AT GR. FLOOR AND FIRST FLOOR SHALL NOT BE CONNECTED TO VERTICAL STACK DISCHARGING AT GR. LEVEL
☐ 4- CONNECTION FROM THE APPLIANCES LINES ON ANY FLOOR AND THE MAIN PIPE SHALL BE IN VERTICAL STAKES ONLY.
☐ 5- FOR ALL BENDS THAT ARE AT BASE OF STACKS, PROVIDE A 45 OR 90 DEGREES LONG RADIUS TYPE ELBOW

VENT PIPES

☐ 1- IN SINGLE STACK SYSTEM, PROVIDE A SEPARATE VENTILATION FOR INDIVIDUAL CONNECTIONS FROM THE WC, FLOOR TRAP, SINK, ETC. ANY HORIZONTAL PIPE LONGER THAN 1.5 METERS FROM SUCH FIXTURES SHALL BE VENTED ON THE HIGH SIDE
☐ 2- VENT PIPES FROM MANHOLES AND VERTICAL STACKS SHALL BE EXTENDED 2 METERS ABOVE THE ROOF, THE END OF WHICH SHALL BE FITTED WITH VENT COWLS.

RESTRICTIONS

☐ 1- THE DRAINAGE PIPES SHALL BE MADE NOT TO RUN THROUGH ELECTRIC ROOMS OR ELECTRIC SUB-STATIONS
☐ 2- DIRECT CONNECTION OF WASTE SYSTEM FROM A FLOOR TRAP TO A FLOOR TRAP SHALL NOT BE PERMITTED.
☐ 3- NO DRAIN PIPES SHALL BE CAST INTO AN RCC STRUCTURAL ELEMENT WITHOUT PRIOR APPROVAL FROM DM STRUCTURAL ENGINEER
☐ 4- TO ALLOW THE DRAIN PIPES TO PASS THROUGH ANY STRUCTURAL ELEMENT IN BUILDING A CAST IRON SLEEVE IS FIRST FITTED WITHIN THE STRUCTURAL ELEMENT SUCH A BY PASS, THE SLEEVE SHALL OFFER A TOLERANCE OF AT LEAST 50 MM FOR EASE OF INSTALLATION OF THE PIPE. THE GAP BETWEEN THE PIPE AND THE SLEEVE SHALL THEN BE FILLED WITH SUITABLE SEALANT.
☐ 5- NOT TO PLACE UNDERGROUND PIPE LINE WITH TOP OF PIPE LESS THAN 600 MM BELOW FFL UNLESS CONCRETE STYLE 150 MM THICK SHALL BE PROVIDED AROUND TO PROTECT THE PIPE

☐ 6- TYPICAL OUTLET SIZE (MM)

WC.	100
BASIN	32
KITCHEN SINK	40
FLOOR DRAIN	75
BATH TUB/SHOWER	40
WASHING M/C	40
BALCONY DRAIN	50

☐ 7- – STACK SIZES SHALL BE AS FOLLOWS:
UP TO GROUND PLUS SEVEN STORY

SOIL PIPE	100
WASTE PIPE	100
VENT PIPE	75
RAIN WATER PIPE	100
BALCONY DRAIN PIPE	50
A/C DRAIN PIPE	32

☐ 8- – STACK SIZES SHALL BE AS FOLLOWS:
ABOVE SEVEN STORY

SOIL PIPE	150
WASTE PIPE	150
VENT PIPE	100
RAIN WATER PIPE	100
BALCONY DRAIN PIPE	50
A/C DRAIN PIPE	32

C) MANHOLES

☐ 1- THE MANHOLE SCHEDULE SHALL BE ARRANGED IN THE MANNER SHOWN BELOW ,THE MANHOLE INVERT LEVEL (I L) , COVER LEVEL (C L) DEPTH AND DISTANCE BETWEEN MANHOLES SHALL BE REFERENCED FROM D.M C , F.I.C USING DM'S DATUM AND ALL UNITS SHALL BE IN IS SYSTEM

MH NO	C L	I L	DISTANCE TO NEXT MH	DEPTH	COVER TYPE	REMARK
D M C						
F I C						
MH NO						

☐ 2- A SKETCH SHOWING THE RELATIVE LEVEL BETWEEN F.I.C AND BUILDING MANHOLES SHALL BE PROVIDED
☐ 3- THE INVERT LEVEL OF THE EXTERNAL DRAINAGE SYSTEM SHALL BE DETERMINED BY THE CONSULTANT TAKING INTO CONSIDERATION THE DRAINAGE CONNECTION LEVEL TO MATCH THAT OF F.I.C, IN CASE OF N.O.C THE DEEP OF LAST MANHOLE SHALL BE LESS THAN 1200 MM.
☐ 4- THE DEPTH OF FIRST MANHOLE SHALL BE AT LEAST 450 MM
☐ 5- SLOPES FOR HORIZONTAL RUNS SHALL BE AS FOLLOWS: FOR 100 MM - 1:60 FOR 150 MM - 1:90 FOR 200 MM - 1:120
☐ 6- MANHOLES AND ACCESS CHAMBERS SHALL BE LAY OUT ACCORDING TO CONSULTANT'S SCHEME., COVERS PROVIDED SHALL BE SUITABLE FOR THE LOADS THEY ARE SUBJECTED TO, AND IN ACCORDANCE WITH BS 497
☐ 7- ACUTE ANGLE CONNECTIONS SHALL NOT BE ALLOWED WITHIN JUNCTION MANHOLES.
☐ 8- AT A MH THE TOPS OF ALL SEWERS SHOULD BE AT THE SAME LEVEL SO THAT THE PIPES OF SMALLER DIAMETER ARE NOT FLOODED WHEN THE BIGGER PIPES ARE RUNNING FULL
☐ 9- BACKDROP SHALL BE PROVIDED WHEN THE LEVEL DIFFERENCE BETWEEN INCOMING AND MAIN SEWER IS CONSIDERABLE
☐ 10- AN INSPECTION CHAMBER / MANHOLE / GULLY TRAP CONSTRUCTED IN COVERED AREA SHALL BE DRY TYPE AND PROVIDED WITH RECESSED DOUBLE SEAL TYPE COVER.
☐ 11- ANY MANHOLE LOCATED IN GARAGE, DRIVEWAY OR OTHER TRAFFIC AREAS SHALL BE PROVIDED WITH HEAVY DUTY COVERS.
☐ 12- IF INSPECTION CHAMBERS / MANHOLES ARE IN AGRICULTURAL LAND, MANHOLE COVERS SHOULD BE RAISED 75 MM ABOVE THE THE NATURAL GROUND LEVEL.
☐ 13- DISTANCE BETWEEN MH TO MH SHALL NOT EXCEED 15 METERS IN VILLAS AND BUILDINGS
☐ 14- MINIMUM REQUIREMENT FOR MANHOLE VENTING SHALL BE THE PROVISION OF VENT PIPE TO FIRST AND LAST MANHOLES OF DRAINAGE LINE. THESE VENT PIPES SHALL BE 100 MM BELOW COVER LEVEL.
☐ 15- THE SITING OF ANY MANHOLE SHALL BE KEPT AWAY FROM UNDERGROUND WATER TANK BY A DISTANCE NOT LESS THAN THE DEPTH OF THE WATER TANK.

SUMP PUMP

☐ 1- IN BASEMENT, SAND TRAP WITH PVC BUCKET SHALL BE PROVIDED JUST BEFORE END CONNECTION OF CARE PARK DRAIN TO THE SUMP PUMP PIT AND THE RAMP DRAIN CHANNEL
☐ 2- IF A BASEMENT SUMP PUMP IS FOR RAIN WATER COLLECTION ONLY FROM BASEMENT, CONNECT ONE WASTE PIPE FROM THE NEAREST TOILET . TO THE BASEMENT DRAINAGE LINE TO MAKE IT ALWAYS WET.
☐ 3- SUMP PUMP SERVICE ONLY THE BASEMENT DRAIN AND SEWERAGE

MANHOLES CONSTRUCTION

☐ 1- ALL INSPECTION CHAMBERS/MANHOLES SHOULD BE BUILT ON BED OF CEMENT CONCRETE 1:4:6 THE THICKNESS OF THE CONCRETE SHALL BE 150 MM FOR MANHOLES UP TO 1000 MM DEEP AND 200 MM FOR DEPTHS ABOVE 1000 MM
☐ 2- WHERE MANHOLE / INSPECTION CHAMBERS ARE CONSTRUCTED BELOW GROUND WATER TABLE , COMPLETE CONSTRUCTION SHALL BE EITHER IN WATERPROOFED RCC. OR IN G.R.P
☐ 3- ALL MAIN LINE CHANNELS SHALL BE LOCATED IN THE CENTER OF THE MANHOLES
☐ 4- BENCHING OF INCOMING BRANCHES SHOULD BE INCLINED TOWARDS THE MAIN DIRECTION OF FLOW .
☐ 5- DIAMETER OF SEMI-CIRCULAR CHANNEL IN THE BOTTOM OF MANHOLE SHALL BE EQUAL TO THAT OF THE INCOMING SEWER DIAMETER..
☐ 6- THE SIDES OF CHANNELS IN MANHOLE SHALL BE EXTENDED VERTICALLY TO THE SAME LEVEL OF THE SOFFIT OF THE PIPE
☐ 7- THE BENCHING IN INSPECTION CHAMBER/ MANHOLES SHOULD HAVE SMOOTH SURFACE CEMENT PLASTER
☐ 8- MANHOLES SIZES SHALL BE AS FOLLOWS:-

MANHOLES DEEP MM	MANHOLES SIZE MM	MANHOLES COVER MM
UP TO 1300	600X600 –	600X600 – FOR SEW FOR SWD
FROM 1301 TO 1700	800X800 –	600X600 – FOR SEW FOR SWD
FROM 1701 TO 2500	1000 ○	600X600 – FOR SEW FOR SWD
FROM 2501 TO 4000	1500 ○	600X600 – FOR SEW FOR SWD

ALL 1000 & 1500 MANHOLE SHALL HAVE G.R.P. LININGS

F I C (THE LAST MANHOLE)

☐ 1- F I C SHALL BE CONSTRUCTED IN CLOSE PROXIMITY WITH THE COMPOUND WALL AND OPPOSITE TO DM CHAMBER/MANHOLE AND THE COVER SHOULD 'BE DUCTILE IRON WITH 6MM THICK G.R.P PUSH FIT SEALING PLATE, THIS MANHOLE SHALL HAVE ONLY ONE INCOMING CONNECTION AND WORK BY GRAVITY ONLY
☐ 2- IN CASE OF THERE ARE NO PROPOSAL OF FUTURE CONNECTION FROM DM DRAINAGE DEP. THE FUTURE CONNECTION SHALL BE IN LESS WIDTH ROAD AND THE MAX DEPTH SHALL BE 1200 MM.

D) MISCELLANEOUS

RAIN WATER DRAINAGE

☐ 1- THE DRAINAGE OF ROOFS AND PAVED AREAS SHALL BE ACCORDING TO BS 6367
☐ 2- RAIN WATER PIPES ARE NOT TO BE CONNECTED TO SEWER LINES. THEY SHALL FREE DISCHARGE ABOVE GROUND
☐ 3- ALL O.T.S, S (4X4 M & LESS) SHOULD HAVE FLOOR TRAPS FOR RAINWATER CONNECTED TO THE NEAREST GULLY TRAP OR WASTE STACK. OTHER O.T.S,S SHALL HAVE RW DRAIN FOR RAIN WATER WHICH IS FREE DISCHARGE TO OUTSIDE.
☐ 4- FOR ALL AIR WELLS, ACCESS DOORS SHOULD BE PROVIDED AT THE LOWER LEVEL OF THE WELL .
☐ 5- FOR PROJECT HAVING LARGE RAIN CATCHMENT AREAS, RAIN WATER SCHEME SHALL BE DESIGNED TO BS 6367 AND DM APPROVAL.

GREASE TRAPS AND OIL INTERCEPTORS

☐ 1- ALL COMMERCIAL KITCHENS AND COMMUNAL ACCOMMODATION KITCHENS SHALL HAVE HEAVY DUTY GREASE TRAPS PROVIDED ACCORDING TO DM REGULATIONS.
☐ 2- FOR VEHICLE WASHING AND SERVICE STATIONS, OIL INTERCEPTORS MUST BE PROVIDED AS PER DM REGULATIONS.

SWIMMING POOLS

☐ 1- DRAINAGE FOR THE SWIMMING POOLS AND BACKWASH PUMPS SHALL BE SHOWN ON SEPARATE DRAINAGE LAYOUTS PLAN. AND THE DRAINAGE LINE SHALL BE SHOWN ON THE RELEVANT DRAINAGE PLAN
☐ 2- FOR SWIMMING POOLS ON THE ROOF OR UPPER FLOORS PROVIDE SEPARATE 2" DIA... PIPE (AFTER PROVIDING REGULATING VALVE)TO. THE GROUND FLOOR MANHOLE CONNECTION .CONNECTION TO BASEMENT SUMP PUMP SHALL NOT BE ALLOWED

PROVISION FOR FUTURE CONNECTION

☐ 1- SHOP FACILITIES IN BUILDINGS SHALL HAVE WP. & VP CONNECTION PROVISIONS FOR FUTURE WITH APPROVED GRATING AS PER DM DETAILS

SAFETY AND HEALTH CONSIDERATIONS:

☐ 1- WINDOW A/C SHALL BE PROVIDED WITH G.R.P. TRAYS HAVING DRAINAGE FACILITY
☐ 2- ALL A/C & BALCONY DRAINS SHALL BE CONNECTED TO WASTE PIPE SYSTEM VIA THE NEAREST FLOOR TRAP OR WASTE STACK.
☐ 3- FLEXIBLE POLYETHYLENE DRAIN PIPES SHALL NOT BE USED FOR A/C CONDENSED DRAIN PIPING
☐ 4- COMMUNAL AND PUBLIC TOILETS IN INDUSTRIAL AND COMMERCIAL ESTABLISHMENT DORMITORIES AND EDUCATION INSTITUTES SHALL BE FURNISHED WITH ASIAN TYPE WC IN PROPORTION NOT LESS THAN 75% OF TOTAL NIMBER OF WC REQUIRED IN ADDITION ,PUBLIC BUILDING SHALL HAVE PROVISION FOR TOILET FACILITIES FOR DISABLED PERSONS
☐ 5- ALL FLOOR DRAIN SHALL HAVE MINIMUM 75 MM DEEP SEAL
☐ 6- FLOOR DRAINS SHALL BE PROVIDED IN ALL GARBAGE ROOMS IN ALL FLOORS AND PUMP ROOMS AND ALL GARAGE ROOMS
☐ 7- ALL WATER SUPPLY PIPES SHALL BE INSTALLED AT LEST 1.0 M AWAY FROM THE SEWER PIPE LINE SEPTIC TANK OR SOAKAWAY.
☐ 8- URINALS SHALL BE CONNECTED TO SOIL PIPES.

SYSTEM TEST

☐ 1- WATER AND AIR TESTS SHALL BE CARRIED OUT AFTER THE PIPES ARE INSTALLED AND CONSULTANT CERTIFICATE SHALL BE CERTIFICATE SHALL BE SUBMITED TO DM ENGINEERS.

E) SEPTIC TANKS, SOAKAWAY AND HOLDING TANKS

☐ 1- SEPTIC TANK AND SOAKAWAY SHALL BE CONSTRUCTED AND SELECTED TO DM DETAILS AND SHALL BE ACCORDING TO BS 6297
☐ 2- THE FOUNDATION LEVEL OF THE BASE OF THE SOAKAWAY SHALL BE KEPT AT LEAST 1 M ABOVE THE WINTER WATER TABLE.
☐ 3- THE FLOOR AREA OF THE SOAKAWAY SHALL BE DETERMINED ACCORDING TO THE 'PERCOLATION RATE IN ACCORDANCE WITH APPROPRIATE TEST IN THE BS 6297
☐ 4- LOCATIONS OF SEPTIC TANK AND SOAKAWAY SHOULD SHOWN ON SETTING OUT PLANS AND ON A GENERAL FOUNDATION LAYOUT PLANS.
☐ 5- SOAKAWAY SHALL BE 1000 MM AWAY FROM THE SEPTIC TANK PLACED AT LEAST.
☐ 6- SEPTIC TANK & SOAK AWAY SHOULD BE CONSTRUCTED AT LEAST 3 MS AWAY FROM ANY FOOTING FOR A BUILDING OR A BOUNDARY WALL AND SHOULD BE CONSTRUCTED FROM SOLID BLOCK WITH A PROVISION OF PROPER INSULATION FOR THE SEPTIC TANK.
☐ 7- IN CASE THERE IS A NEED TO CONSTRUCT THESE TANKS AT A DISTANCE LESS THAN 3. MS AND MORE THAN 1 MS FROM ADJACENT FOOTINGS THE SEPTIC TANK SHOULD BE STRUCTURES FROM REINFORCED CONCRETE WITH A PROVISION OF PROPER INSULATION TO PREVENT ANY LEAKAGE FROM THE TANK
☐ 8- IF THE SEPTIC TANK & SOAK AWAY ARE TO BE FOUNDED AT A LOWER LEVEL THAN THAT OF ADJACENT FOOTING THEY SHOULD BE CONSTRUCTED FIRST.
☐ 9- IF SEPTIC TANK CONSTRUCTED FROM FIBERGLASS MATERIAL THE DIMENSION SHALL BE THE SAME ,THE THICKNESS SHALL NOT LESS THAN 6 MM AND IT SHOULD BE INCASED WITH PROPER PROTECTION SOLID BLOCKS .
☐ 10- PROPER INSULATION SHOULD BE PROVIDED FOR FOUNDATIONS OF BUILDINGS IN SUCH CIRCUMSTANCES FOR PROTECTION AGAINST SULFATES.
☐ 11- SOAKAWAY SHOULD BE FILLED WITH BOULDERS OF SIZE 75 MM TO 100 MM
☐ 12- GRP SEALING PLATE SHALL BE PROVIDED FOR THE LAST MANHOLE BEFORE THE SEPTIC TANK AND THE MANHOLE FOR THE FUTURE CONNECTION BEFORE THE DM SEWER LINE
☐ 13- SIDE LEAKAGE FROM SOAK AWAY SHOULD BE PREVENTED OF CONSTRUCTED AT THE SAME LEVEL OF ADJACENT FOOTING OF A BUILDING OR A BOUNDARY WALL

ENGINEERING CONSULTANT.

NAME : ENG:
SIGNATURE :
DATE :

STAMP

This 2D drawing is extracted from 3D BIM Model

Key Plan

N

General Notes :

01. Do not Scale The Drawings. Follow (The MM) Written Dimensions Unless Otherwise Mentioned.
02. Each Drawing Shall be Read in Conjunction With all Other Drawings,Specification & the General Standard Details. In Case of Any Discrepancies Bring it to the Knowledge of Engineer and Same to be Resolved Prior to Execution of Work.
03. Dimensions Shown on These Drawings are Structural Dimensions. Unless Noted Otherwise.
04. All Levels Indicated On These Drawings are Levels as per the the Survey Datum, Same Should be Verified at Site Prior to Execution of Work.
05. All Levels Indicated On These Drawings are Based On Design Level, Which is Equivalent to Datum Level and Same Should be Verified at Site Prior to Execution of Work.

AB TENDER DRAWINGS

No.	Date	Initials	Description
00	10/03/2026	LSG	TENDER SUBMISSION
00	16/01/2026	LSG	DETAILED DESIGN SUBMISSION
00	31/10/2025	LSG	SCHEMATIC DESIGN SUBMISSION
00	15/10/2025	LSG	SCHEMATIC SUBMISSION-50%
00	01/09/2025	LSG	CONCEPT DESIGN SUBMISSION

REVISIONS / ISSUE

THIS DRAWING MUST NOT BE REPRODUCED IN WHOLE OR IN PART WITHOUT THE PRIOR APPROVAL OF ARIF & BINTOAK

Status
TENDER SUBMISSION

Client
EMAAR

Lead Consultant
الأستشاري:
عارف وبن طوق
Arif&Bintoak
consulting architects & engineers
POST BOX 3670 DUBAI.UAE TEL 04-3372888 FAX 04-3370352
POST BOX 42981 ABU DHABI.UAE TEL 02-634669 -artoak@emirates.net.ae

Project Title:
Proposed Residential Townhouses
and Standalone Villas
TH-1, TH-2, and TH-3 at Grand Polo
Club & Resort, Dubai, UAE

Discipline : MECHANICAL

Drawing title:
DRAINAGE NOTES

Architectural Engineer	Electrical Engineer	Structural Engineer
MUSTAFA ABID	MARWAN MOHD.	RIHAM RAMADAN
Registration # 5847	Registration # 80102	Registration # 105092
Scale(s): NTS	Date : 10/03/2026	Project no: 1857
Drawn by: LSG	Checked : MAV	Approved : MRM

Drawing Number : GPCR-TH01-AB-DRG-ZZZ-MEC-DR-3002-00

16-Feb-26 2:59:10 PM

© AB I A1 Title Block . BIM Standard Ver. 2.0